

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Kelvin Waves

Lab 2

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Questions

Theory

Coastal Kelvin Waves

1. Describe in your own words what coastal Kelvin waves are.
2. State the necessary assumptions to study coastal Kelvin waves.
3. Write down the solutions for the Shallow Water Equations for coastal Kelvin waves.
4. Describe the shape of the coastal Kelvin waves.

Equatorial Kelvin Waves

1. Describe the assumptions to study equatorial Kelvin waves.
2. Write down the solution for the Shallow Water Equations for equatorial Kelvin waves.
3. In which direction do the equatorial Kelvin waves travel?

Experiments

Set up the model as described in the laboratory description for Laboratory 2 and follow the instruction for the experiments.

Coastal Kelvin Waves

1. Compute the phase speed from the position of the peak of the wave along the x-axis at $y = 0$ and at different times.
2. Compare your results to the theory and discuss weaknesses of your results.
3. Provide plots showing the cross-section in the y-direction of your simulated coastal Kelvin waves. Ensure to include the analytical solution and the initial condition.
4. How is the result dependent on the resolution?
5. Show plots of the cross section of these waves in x-direction for your experiments at different time steps.
6. Discuss changes of the wave with time and with resolution.

Equatorial Kelvin Waves

1. Why do we want reduced gravity in this simulation?
2. Provide a plot of the horizontal cross-section at an appropriate time step and discuss the waves you can identify (include a cross-section through the equatorial Rossby waves in the same plot).
3. Provide a plot of the wave for multiple points in time (in x-direction, at $y = 0$).
4. From that, compute the phase speeds and compare to theory.
5. Provide a plot showing cross sections of the wave amplitudes in y-direction. Make sure to also include the initial condition and the analytical solution and compare your result to the theory.
6. Provide examples of where these waves are observed in the real ocean.

1 Theory

1.1 Coastal Kelvin Waves.

(i) Kelvin waves are a geostrophic phenomenon where the presence of a coast forces an initial perturbation to propagate along that coast. The movement is anticlockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere.

The reason for these waves propagating the way they do lies in the presence of the boundary (like a coast or the equator). In normal geostrophic balance a large-scale high or low (same as a 'bump' or 'valley') can exist due to geostrophic movement around the high/low, maintaining the shape due to the presence of the Coriolis force.

But if a boundary is present, no such flow around the high/low can exist. If one for example has a high, normal geostrophy tells us that we need anticyclonic motion around the high to keep the 'bump' in place. However, if we were to have a southern boundary (and we assume that we are in the Northern Hemisphere), then we would lack westerly geostrophic flow. Thus, due to the conservation of momentum, this high would have to propagate to the East. That is, in other words, a Kelvin wave.

(ii) Equations. B.C.

iii Write solh

2 Experiments

2.1 Coastal Kelvin Waves

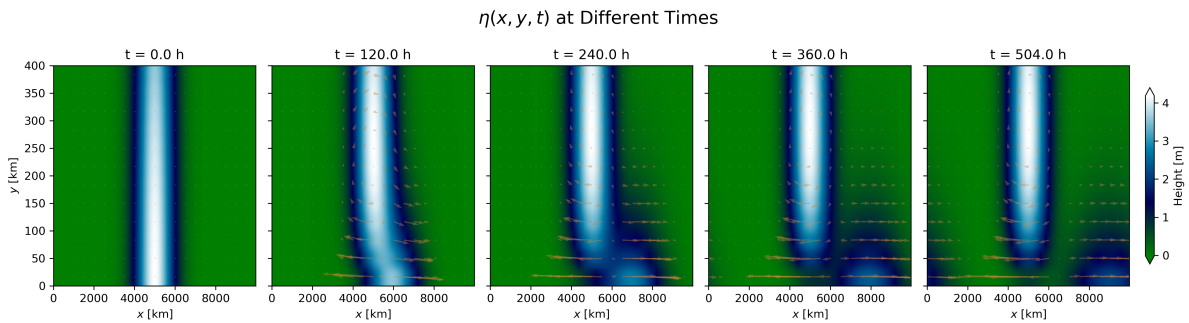


Figure 1: This figure...

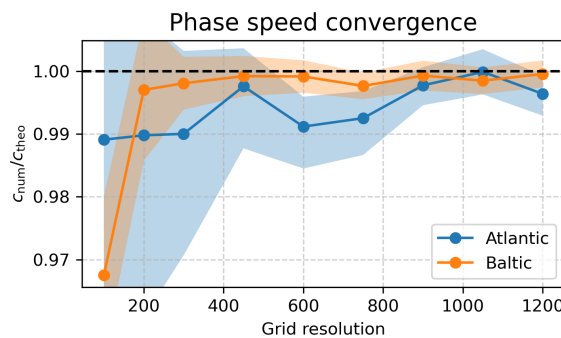
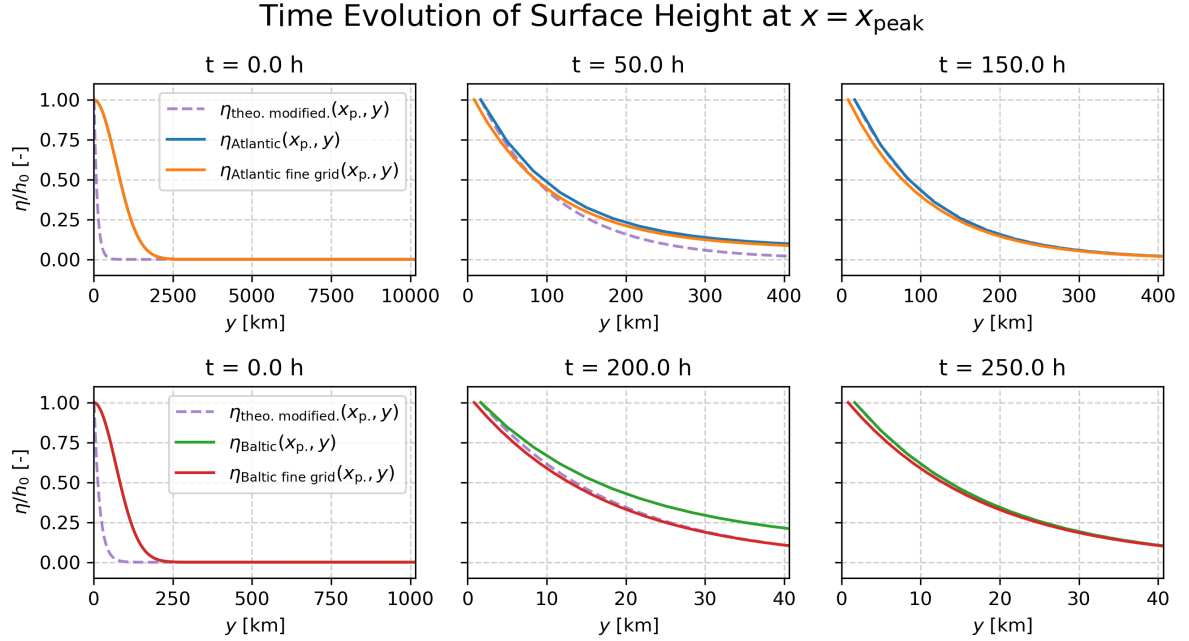
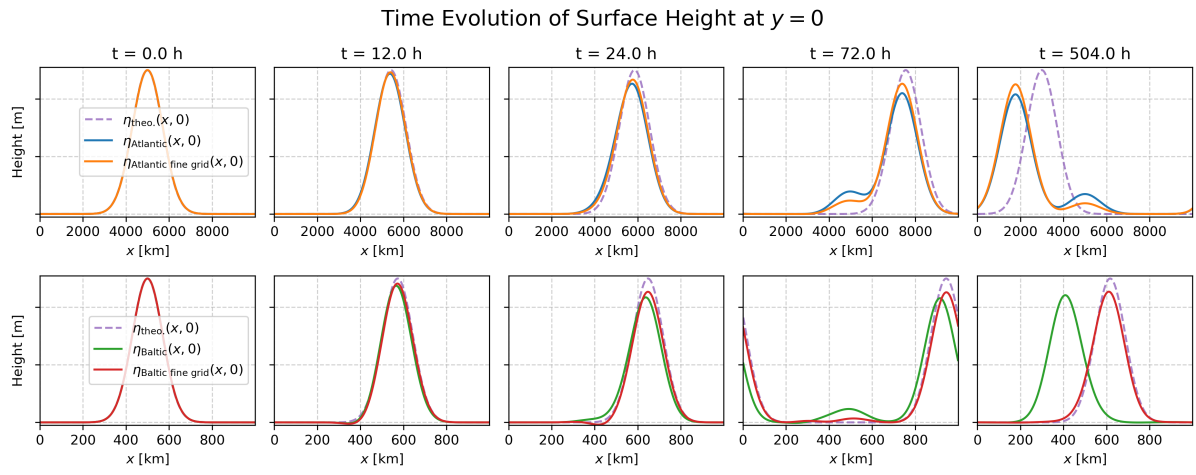


Figure 2: This figure...

Case	Time (h)	c_{num}	c_{theo}	Rel. error (%)
Atlantic	0 $\rightarrow$ 1	12.8646	9.9045	29.89
	0 $\rightarrow$ 10	10.7813	9.9045	8.85
	0 $\rightarrow$ 24	11.4179	9.9045	15.28
	0 $\rightarrow$ 168	10.5912	9.9045	6.93
	0 $\rightarrow$ 336	10.6256	9.9045	7.28
	0 $\rightarrow$ 504	10.5728	9.9045	6.75
Atlantic fine grid	0 $\rightarrow$ 1	16.3367	9.9045	64.94
	0 $\rightarrow$ 10	10.2026	9.9045	3.01
	0 $\rightarrow$ 24	10.7910	9.9045	8.95
	0 $\rightarrow$ 168	10.5705	9.9045	6.72
	0 $\rightarrow$ 336	10.5877	9.9045	6.90
	0 $\rightarrow$ 504	10.5705	9.9045	6.72
Baltic	0 $\rightarrow$ 1	2.7366	1.7155	59.52
	0 $\rightarrow$ 10	1.7875	1.7155	4.20
	0 $\rightarrow$ 24	1.8203	1.7155	6.11
	0 $\rightarrow$ 168	1.8258	1.7155	6.43
	0 $\rightarrow$ 336	1.8279	1.7155	6.55
	0 $\rightarrow$ 504	1.8304	1.7155	6.70
Baltic fine grid	0 $\rightarrow$ 1	3.0838	1.7155	79.76
	0 $\rightarrow$ 10	1.6834	1.7155	1.88
	0 $\rightarrow$ 24	1.7190	1.7155	0.21
	0 $\rightarrow$ 168	1.7190	1.7155	0.21
	0 $\rightarrow$ 336	1.7187	1.7155	0.19
	0 $\rightarrow$ 504	1.7177	1.7155	0.12

Table 1: Kelvin wave phase speed: numerical vs theoretical comparison.**Figure 3:** This figure...

**Figure 4:** This figure...

2.2 Equatorial Kelvin Waves

(i)

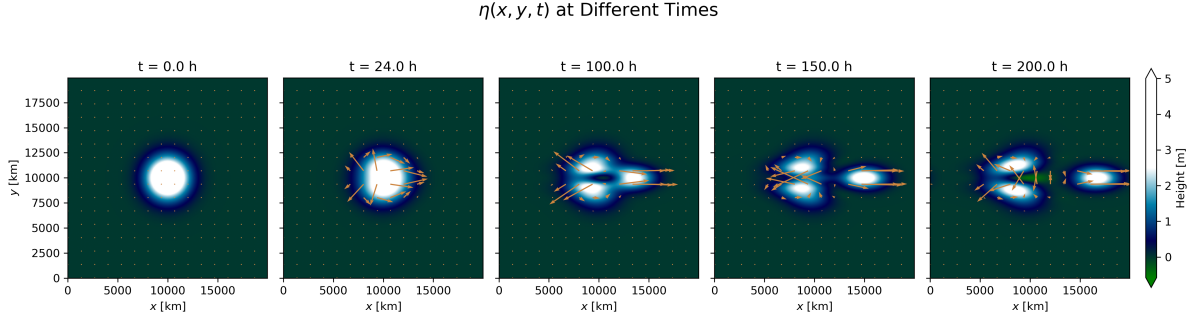


Figure 5: This figure...

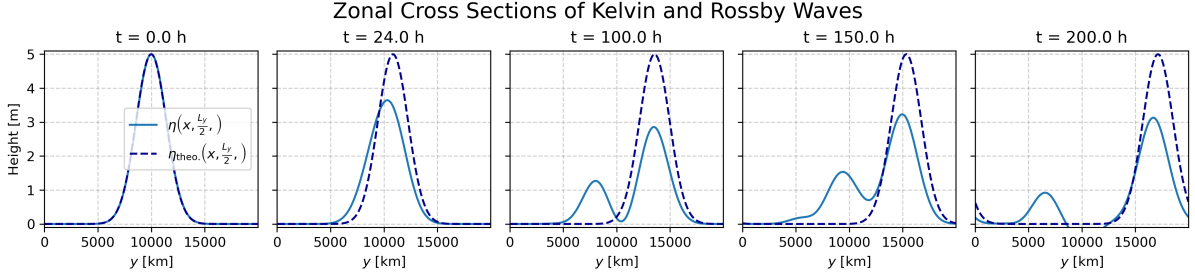


Figure 6: This figure...

Time [h]	$x_{\max}$ [km]	c_{num} [m/s]	c_{theo} [m/s]	Error [%]
0 → 24	10300.0	3.86	9.90	61.05
24 → 100	13500.0	11.70	9.90	18.09
100 → 150	14966.7	8.15	9.90	17.73
150 → 200	16700.0	9.63	9.90	2.78
0 → 200	16700.0	9.33	9.90	5.81

Table 2: Kelvin-wave phase speed diagnostics taken from Figure 6.

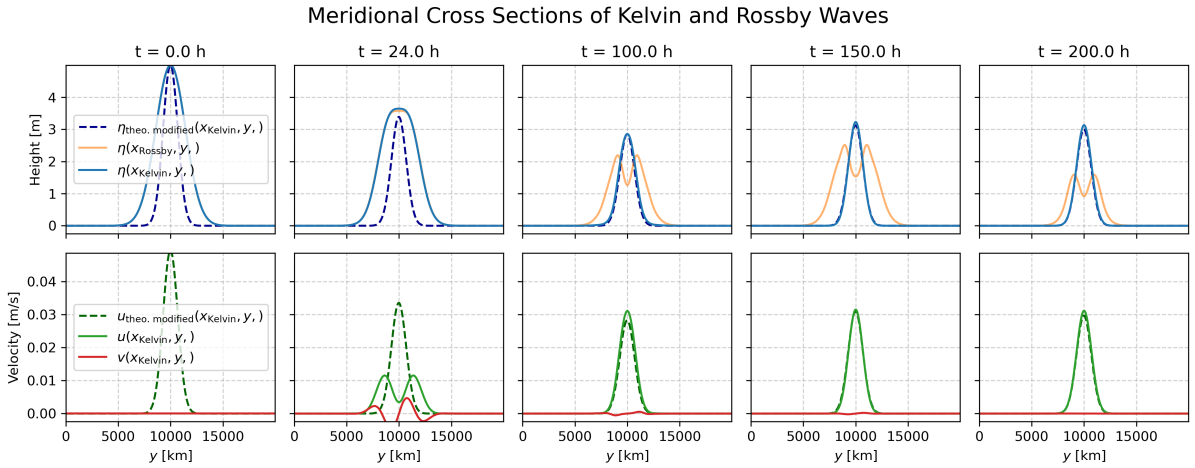


Figure 7: This figure...

A Appendix

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/
Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`